

10th International Conference of Information and Communication Technology (ICICT-2020)

Retrieval of rainstorm similarity system based on deep learning

Weiping Zhang^{a,b,*}, Cunming Hao^{a,b}, Zhiming Hu^c

^aHebei Academy of Sciences Institute of Applied Mathematics, Shijiazhuang, Hebei, 050081, China

^bInformation Security Certification Engineering Technology Research Center, Shijiazhuang, Hebei, 050081, China

^cChina Railway Shanghai Design Institute Group CO., LTD, Shanghai, 200070, China

Abstract

Weather map contains abundant meteorological information. The analysis of high-altitude low-pressure vortex plays a crucial role in rainstorm weather forecast. Using the graphical and numerical information of the low-pressure vortex, the similar weather system in the historical storm database can be searched for the accurate prediction of weather conditions. A deep learning convolutional neural network and YOLO V3 algorithm are studied. A similarity search algorithm for weather systems based on deep learning and high-altitude low-pressure vortex is designed. The method first pre-processes the height field meteorological data and generate a height field contour image; Then use YOLO V3 algorithm to locate and identify the vortex in the image and superimpose the wind field to remove the high-voltage vortex. Finally, the center position, center strength, system intensity, and influence region information of each low-pressure vortex are calculated and used for similar retrieval of weather systems. The experimental results show that the method works well and it provides a great help for the accurate prediction of heavy rain.

© 2021 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: Deep learning, convolution neural networks, high-altitude low-pressure vortex, meteorological disasters, YOLO V3;

1. Introduction

Rainstorm weather is a high-impact meteorological disaster. Accurate and timely forecast can effectively prevent meteorological disasters, reduce casualties and property losses. At present, a lot of weather forecast work at home and abroad focuses on analyzing the occurrence of disastrous weather by using conventional weather charts and numerical forecast products. Forecasters mainly rely on experience to forecast. The experience of forecasters mainly

* Corresponding author. Tel.: 0311--67306779.

E-mail address: 872863057@qq.com

comes from the accumulation and synthesis of typical weather systems, and the forecast conclusion largely depends on the similar search of historical data. Therefore, the key problem of weather forecast is how to quickly and accurately retrieve the weather system similar to the current weather situation from a large number of historical databases. Lian zhiluan¹ et al. tried to use a similar-difference method to make temperature forecast in villages and towns, and achieved preliminary results; Liu Yong² et al. studied graph similar retrieval based on isoline analysis technology; Wang Ping³ et al. studied similar retrieval method of weather map.

Similar retrieval of weather map needs to use image analysis technology to extract key features of weather map. Image analysis technology based on deep learning is in the ascendant. In recent years, deep learning theory has developed rapidly. Different from the traditional feature extraction algorithm which relies on prior knowledge, deep neural network can construct feature description adaptively driven by training data, which has higher flexibility and universality. As an important technology to realize deep learning, convolutional neural network (CNN)⁴ has successfully trained the first deep neural network structure, and has made outstanding achievements in image recognition^{5,6,7}, speech recognition⁸ and other fields. In the field of meteorology, in-depth learning also has preliminary application. Lu Zhiying⁹ and other researchers used DBNs to study the automatic recognition model of short-term heavy rainfall, and Yang Han¹⁰ and Liu Xinda¹¹ used in-depth learning technology to study the refined prediction of temperature. Although deep learning technology has been initially applied in the field of meteorology, it has not been applied in the field of low pressure vortex recognition.

In the high altitude weather map data, the area near the low pressure vortex in the height field is a rainy area. The accurate location and analysis of the low pressure vortex are the prerequisites and conditions for accurate forecasting of the rainstorm weather. Through the analysis of the rainstorm weather map in Hebei Province in the past 20 years, the characteristics of the low pressure vortex in the weather map are extracted by using the depth learning image analysis technology, and the weather map similar to the target image is searched in the historical database. It is helpful for the accurate forecast of rainstorm.

2. Convolutional Neural Network (CNN)

Inspired by the structure of biological neural network, convolutional neural network truly realizes the learning and training of multi-layer network structure for the first time. It is composed of multiple convolution layers, pooling layers (also known as the lower sampling layer) and a complete fully connection layer. The convolution layer obtains the local characteristic map of the image through the filter, and its weight sharing and the characteristics of the local receptive field greatly reduce the network parameters; the pooling layer reduces the resolution of the characteristic map to make the image highly resistant to spatial deformation such as translation, amplification and contraction, rotation¹².

Convolution layer For the input image X of size $m \times n$, a group of trainable filter banks of size $w \times w$ are used to convolute the input image X , the moving step of the convolution filter is s . The convolution characteristic image obtained is Y , and the size is $((m-w)/s+1) \times ((n-w)/s+1)$. The convolution process is as follow:

$$y_i = f\left(b_i + \sum_j k_{ij} * x_i\right) \quad (1)$$

Among them, $*$ is the convolution operation, x_i is the input graph, y_i is the output graph, k_{ij} is the filter parameter, b_i is offset and f is activation function.

- 1) Pooling layer. In the convolution neural network, the pooling layer is usually behind the convolution layer. By pooling, the dimension of image eigenvector output from the convolution layer can be greatly reduced. Firstly, the convolution characteristic map should be divided into several small areas (overlapping partition can be carried out), and then the characteristics of each area should be aggregated by pooling method to get a new characteristic map. This pooling process can be average pooling, maximum pooling, random pooling¹³ etc.
- 2) Fully connected layer. One or more fully connected layers are connected after the convolution layer and the

pooling layer. The convolution layer and the pooling layer are used to extract the features of the input image and feed them back to the fully connected layer for feature classification. Therefore, the fully connected layers play the role of "Classifier" in the whole convolution neural network.

3. High Altitude Low Pressure Vortex Recognition Based on CNN

The traditional low-pressure vortex recognition methods mainly include heuristic search A * algorithm¹⁴, DP algorithm¹⁵, etc. Although A * algorithm has certain effect on recognition, the algorithm generally uses a lot of space to store the searched intermediate state. To prevent repeated search, the algorithm needs a lot of storage space to store the searched information; DP algorithm has a good effect on identifying the region of low pressure vortex with large gradient. However, for areas with sparse isobars, the large distance between the feature points in this area can not satisfy the requirements of distance factor, so this algorithm will lead to data loss in this area.

Yolo¹⁶ series is another series of excellent detection algorithms after fasterr-cnn. The fasterr-cnn series of algorithms take the steps of "candidate frame extraction + classification" (i.e. RPN + fastcrnn) to achieve multi-target detection. Its essence is to transform the detection problem into a classification problem with high accuracy, but the speed is not ideal. Using a joint training method of target classification and detection, Yolo directly regresses the location and category of the target frame in the output layer, and transforms the detection problem into a regression problem, which greatly improves the detection speed. Yolo V3 adds multi-scale prediction based on Yolo v2. The classifier uses multiple logistic classifiers to classify each frame. Yolo V3 is a full convolution neural network, which uses a large number of residual jump layer connections. Its core network framework is the feature extraction network darknet-53 with 53 layer convolution layer, including 52 layer convolution layer and 1 full connection layer. It adds multiple consecutive convolutions of 3×3 and 1×1 to increase direct connection. Therefore, this paper uses the convolution neural network algorithm based on Yolo V3 to identify the high-altitude vortex automatically.

According to the detection results of the Yolo V3 network, four characteristics of the vortex in the target area, including the center position, the center strength, the system strength and the influence area, can be got. The center position of the vortex is defined by the outer center of the inner circle, the center strength is defined as the value of the isoline of the inner circle, and the system strength C is the ratio of the value difference of the closed isoline around the center to the number of isolines. Since the identified vortex is in a rectangular frame, the influence area can be represented by the area s of the rectangular frame. Figure 1 shows the flow chart of vortex recognition based on convolution neural network. Figure 2 shows the vortex recognition result by Yolo V3.

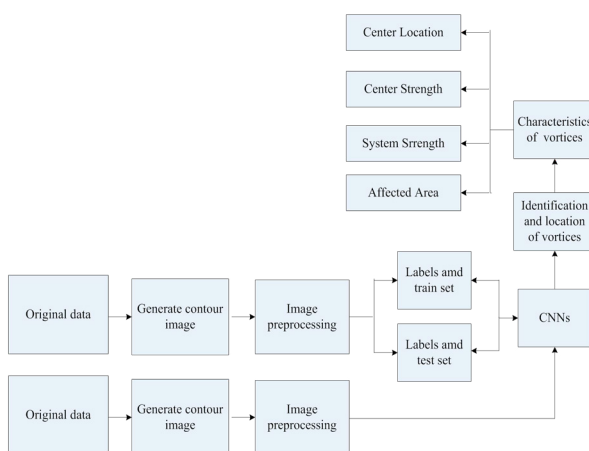


Fig. 1. Vortex Identification Flow Chart Based on CNNs

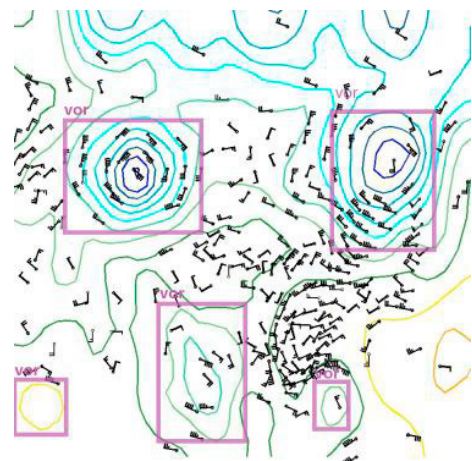


Fig. 2. The Vortex Recognition Result by Yolo V3

4. Retrieval of similar weather map based on low pressure vortex

Similar weather map retrieval is to find out the most similar case in a large number of historical weather maps. The retrieval of similar weather maps based on low pressure vortices is determined by the follow factors, such as center position, center strength, system strength and influence area of low pressure vortices, etc.

4.1. Determination of low pressure vortex based on wind field

Vortices are composed of closed isolines. The characteristics of vortices are quite obvious in the weather map. However, the identification of high pressure vortices and low pressure vortices needs to be carried out according to the change of wind field. Therefore, Yolo V3 convolution neural network can only find all vortices, and can not identify whether they are low pressure vortices or not. When the wind field is superimposed on the contour image, the low pressure vortex can be determined according to the change of wind direction. The low vortex should have cyclonic rotation conditions. The judgment method is: the sign of the sum of the projections of the wind of the closed low vortex contour in the anticlockwise tangential direction of the contour should be positive¹⁷; using the judgment condition to make a secondary judgment on the identified vortex, the low-pressure vortex that meets the cyclonic conditions is the low-pressure vortex. Figure 3 shows the low pressure vortex recognition result based on wind field.

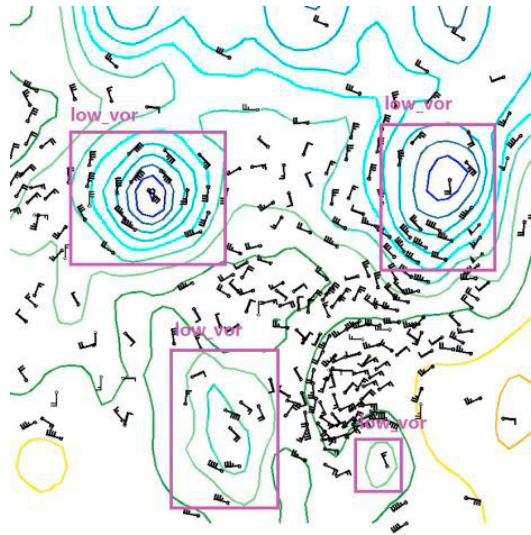


Fig. 3. Low Pressure Vortex Recognition Result based on wind field

4.2. Weighted Similar Search Algorithm

There are four main characteristics of low-pressure vortices. The similarity retrieval of the weather map is based on the center position, center strength, system strength and the priority of the affected area, and the similarity of low-pressure vortices is calculated with the weighted similar search algorithm. It is assumed that there are multiple low-pressure vortices on a real-time weather map, and there are n maps in the historical database, and there are n_i low-pressure vortices on each map and $i = 1, 2, \dots, n$. Similar retrieval methods and processes are as follows:

(1) Central location similarity search

Let the center of the k -th low pressure vortex on the real-time weather map be $z_k(x_k, y_k)$, The j -th low-pressure vortex center of the p -th historical map most similar to the real-time weather map is $z_{pj}(x_{pj}, y_{pj})$, $j = 1, 2, \dots, n_i$, The distance between the centers of low pressure vortices is:

$$d_{pk} = \min \left\{ \sqrt{(x_k - x_{pj})^2 + (y_k - y_{pj})^2} \right\} \quad (2)$$

The distance threshold value is T_d . if $d_{pk} \leq T_d$, continue to calculate, otherwise it is considered not similar. According to formula (2), the image in all the historical databases can be processed to get the closest historical weather map library to z_k position. By traversing the n_i low-pressure vortex centers on the real-time weather map, we can get the most similar historical chart library Ω_d .

The similarity calculation formula of central location distance is:

$$r_d = 1 - \frac{d_{pk}}{T_d} \quad (3)$$

It can be seen from the above formula that, when the distance threshold value T_d is given, the larger r_d is, the more similar the central position of the low-pressure vortices is.

(2) Center strength, system strength and influence area similarity search

Let the center strength value of the center z_k of the k-th low pressure vortex on the real-time weather map be v_k , The j-th low-pressure vortex center strength of the center z_{pj} of the p-th historical map most similar to the real-time weather map is v_{pj} , $j=1,2,\dots,n_i$, The center strength difference is:

$$V_{pk} = \min |v_k - v_{pj}| \quad (4)$$

The center strength difference threshold value is T_v . if $V_{pk} \leq T_v$, continue to calculate, otherwise it is considered not similar. By processing all the images in Ω_d according to the similarity retrieval of the center position, we can get the Historical Map Library Ω_v , which is similar to the center position and the center intensity of the low-pressure vortex in the real-time weather map.

The calculation formula of center intensity similarity is:

$$r_v = 1 - \frac{V_{pk}}{T_v} \quad (5)$$

It can be seen from the above formula that, when the center strength difference threshold value is T_v is given, the larger r_v is, the more similar the central strength of the low-pressure vortices is.

At this time, the similarity retrieval has considered the joint influence of center position and center strength. The similarity retrieval method and the similarity calculation formula for the system strength and the influence area are basically similar to the center strength, so when the threshold value is given, the history library Ω_c with similar system strength and the History Library Ω_s with similar influence area can be obtained.

Weighted similar search algorithm

By adjusting the size of each threshold, we can ensure that the set Ω_d , Ω_v , Ω_c , Ω_s exist, not empty and satisfy the relational expression $\Omega_s \subseteq \Omega_c \subseteq \Omega_v \subseteq \Omega_d$, so the set Ω_s can be regarded as the final similar historical map library. Finally, the weather similarity calculation formula based on low pressure vortex is as follows:

$$r = \omega_d r_d + \omega_v r_v + \omega_c r_c + \omega_s r_s \quad (6)$$

Among them, ω_d , ω_v , ω_c and ω_s are the weights seted by forecasters according to their own experience, and these weights satisfy the follow relationships:

$$\begin{cases} 0 \leq \omega_i \leq 1 \\ \sum_i \omega_i = 1, i = d, v, c, s \end{cases} \quad (7)$$

From formula (6), the similarity r of all similar historical graphs in Ω_s can be given. The greater r is, the higher the similarity is. According to the similarity calculation results, we can sort the images similarity in Ω_s . According to this sort, forecasters can quickly retrieve the most similar historical cases from the historical weather map database.

4.3. Experiment

4.3.1. Experimental data

The data set is the rainstorm weather data from 1999 to 2017 obtained from a City Meteorological Bureau. The original data is the fourth kind of data saved by MICAPS, which is the spatial grid numerical data. Since the input of convolution neural network is image, it is necessary to preprocess the collected data to generate contour image. The number of grid data saved in different years is different, so it is necessary to normalize the image. The image size after normalization is 832 * 832. Due to the lack of rainstorm data, the small data set is not enough to train the network. Therefore, we process the original data set by enhancing. The size of the enhanced data set is 11440, including 10296 images for network training and 1144 images for testing.

After preprocessing the collected data set, the historical rainfall database of the city is established to facilitate future query and retrieval.

4.3.2. Experimental result

The given similar search area specified in this paper is $75^\circ\text{E}-145^\circ\text{E}$, $20^\circ\text{N}-70^\circ\text{N}$. Combined with the experience of three forecasters from the Municipal Meteorological Bureau and the results of many experiments, set weights to $(\omega_d, \omega_v, \omega_c, \omega_s) = (0.4, 0.2, 0.2, 0.2)$, set threshold to $(T_d, T_v, T_c, T_s) = (10, 12, 0.2, 10)$. The retrieval system is embedded in MICAPS for development, which not only improves the accuracy of prediction, but also facilitates the operation of forecasters.

Figure 4 shows the 500 hPa geopotential height field at 20:00 on June 24, 2013. Figure 5, Figure 6 and Figure 7 are used to retrieve the top three 500 hPa geopotential height field most similar to Figure 4 from the historical database.

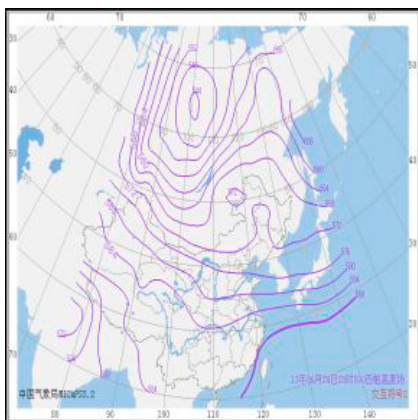


Fig. 4. 500hPa Height Map of 2013062420

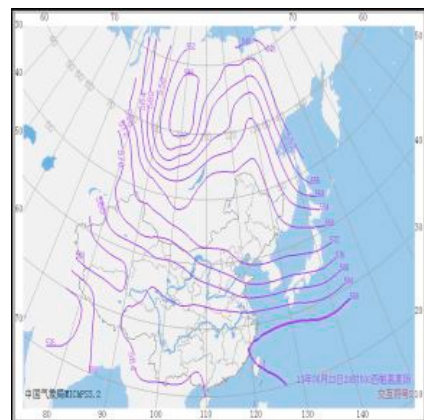


Fig. 5. 500hPa Height Map of 2013062320

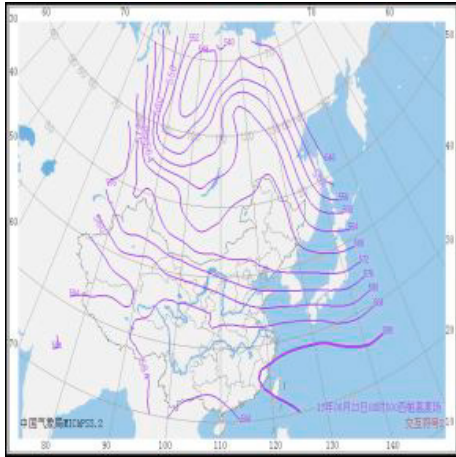


Fig. 6. 500hPa Height Map of 2013062308

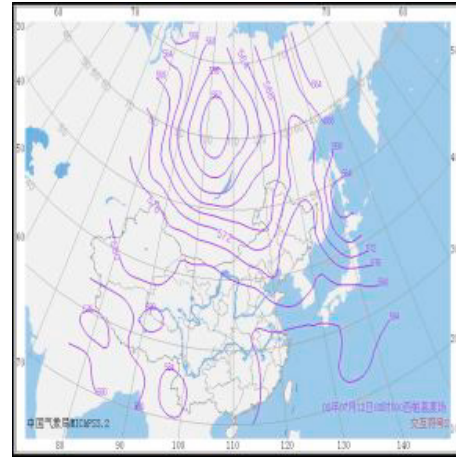


Fig. 7. 500hPa Height Map of 2008071208

It can be seen from the experimental results that the similarity retrieval algorithm based on deep learning and high-altitude low-pressure vortices has a good effect on the retrieval of similar weather map of rainstorm, which greatly reduces the calculation time and computer storage space in the identification and location of vortices; greatly improves the accuracy of rainstorm weather forecast through the accurate identification and location of high-altitude low-pressure vortices; and is similar to the MICAPS platform In combination, not only the forecaster's proficiency in the operating system is taken care of, but also the MICAPS data can be called at any time to update the database in time.

5. Conclusion

In this paper, deep learning convolution neural network is used to identify the center position, center strength, system strength and influence area of the high-altitude low-pressure vortex, and then the similarity retrieval algorithm of rainstorm weather based on deep learning and high-altitude low-pressure vortex is studied. The established historical rainstorm database is stored on the rear server, which provides convenience for the promotion and use of the system. The algorithm in this paper improves the accuracy of prediction. Next, we will continue to study weather systems such as high trough line, ridge line and shear line, etc.

Reference

1. Lian Zhiluan, Li Guocui, Bian Tao. Making temperature forecast of villages and towns based on multi-level similarity difference method[J]. Meteorological, 2008,34(5): 113-117.
2. Liu Yong, Zhou Xiaolin, Wu Biwen. Image similarity retrieval based on isoline analysis technology[J]. Computer application, 2006, 26: 5-7.
3. Wang Ping, Kong Xiumei, Yang Hongmin. Research on similar retrieval of weather map[J]. Journal of Tianjin University, 2004, 37(3): 264-268.
4. Li Yandong, Hao Zongbo, Lei Hang. Survey of convolutional neural network[J]. Journal of Computer Applications, 2016, 36(9): 2508-2515.
5. Lecun Y, Boser L, Denker JS, et al. Backpropagation applied to handwritten zip code Recognition[J]. Neural Computation, 1989, 1(4): 541-551.
6. Szegedy C, Liu W, Jia YQ, et al. Going deeper with convolutions[C]. In Proc. of the IEEE Conf on Computer Vision and Pattern Recognition(CVPR), 2015, 1-9.
7. Simonyan K, Zisserman A. Very Deep Convolutional Networks for Large-Scale Image Recognition. In Proc[C]. International Conference on Learning Representations(ICLR). 2015.
8. Han K, Wang D L. Neural network based pitch tracking in very noisy speech[J]. IEEE/ACM Transactions on Audio, Speech, & Language Processing, 2014, 22(12): 2158-2168.
9. Lu Zhiying, Ren Yimo, Sun Xiaolei. Weather recognition system of short-term heavy rainfall based on deep learning[J]. Journal of Tianjin University(Natural science and Engineering Technology Edition). 2008, 51(2): 111-119.
10. Yang Han. Research on meteorological prediction based on deep learning[D]. Ha erbin:Harbin Institute of Technology, 2017.

11. Liu Xinda. Research on Meteorological temperature prediction based on deep learning[D]. Yinchuan: Ningxia University, 2016.
12. ZHOU Feiyan, JIN Linpeng, DONG Jun. Review of convolutional neural network[J]. Chinese Journal of Computers, 2017, 40(6): 1229-1251.
13. Y-Lan B, Jean P, Yann L C. A Theoretical Analysis of Feature Pooling in Visual Recognition[C]. In: Proceedings of the 27th International Conference on Machine Learning, Haifa, Israel, 2010: 371–382.
14. Zhang Xiaowei. Recognition and application of wind shear line based on MICAPS platform[D]. Wuhan: Wuhan University of Technology, 2016.
15. Dai Xi, Li Qian, Gu Daquan. Automatic identification of slot line based on DP algorithm[J]. Meteorological Technology, 2016, 44(1): 36-40.
16. Redmon J, Divvala S, Girshick R, et al. You Only Look Once: Unified, Real-Time Object Detection[J]. Computer Science, 2016: 779-788.
17. Lin Zhiqiang, Zhou Zhenbo, Jia La. Objective recognition method of Plateau vortex and its preliminary application[J]. Plateau Meteorology, 2013, 32(6):1580-1588.